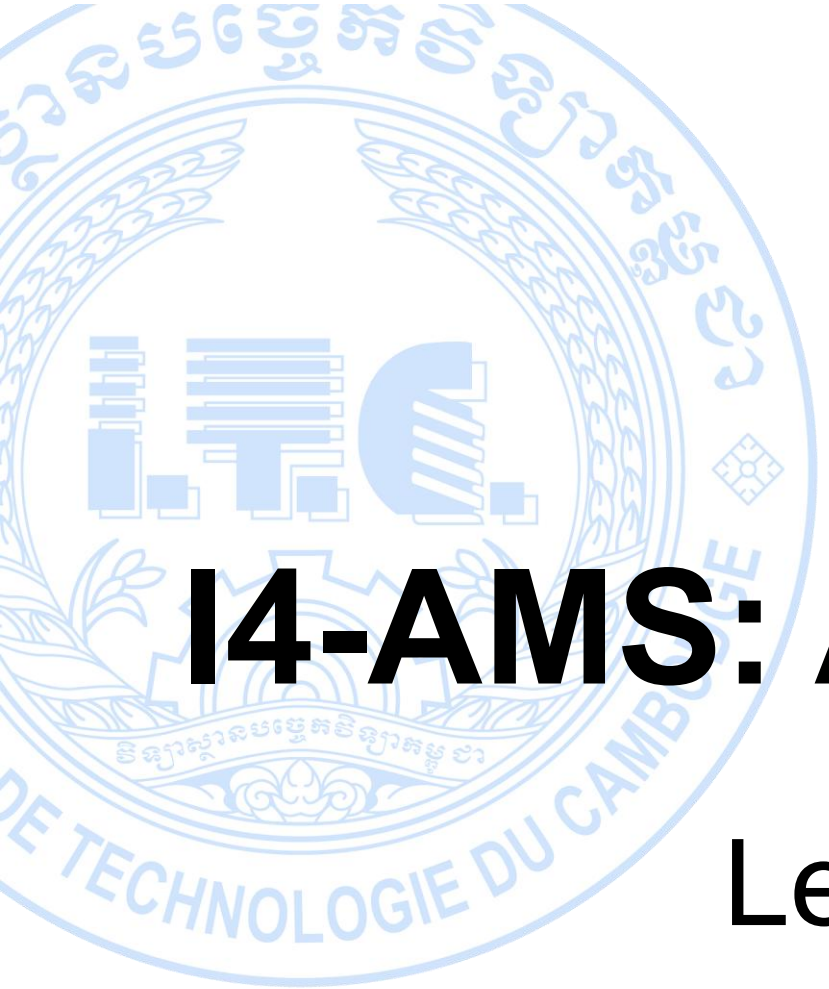


I4-AMS: Artificial Intelligence

Lecture 1: Introduction

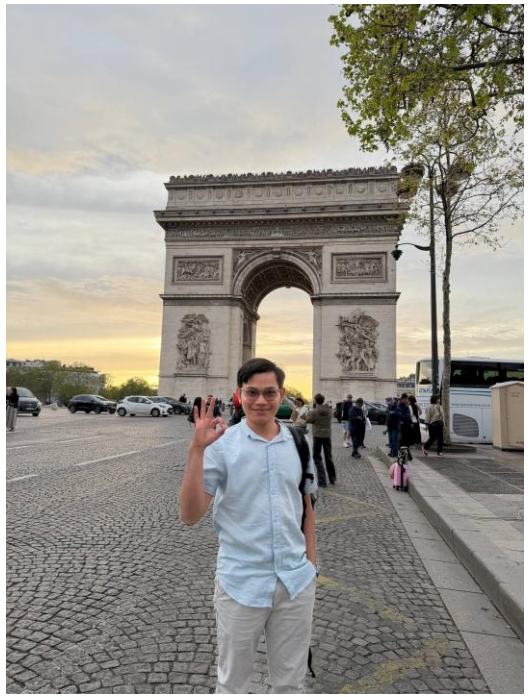


Welcome to Artificial Intelligence Class

Instructors



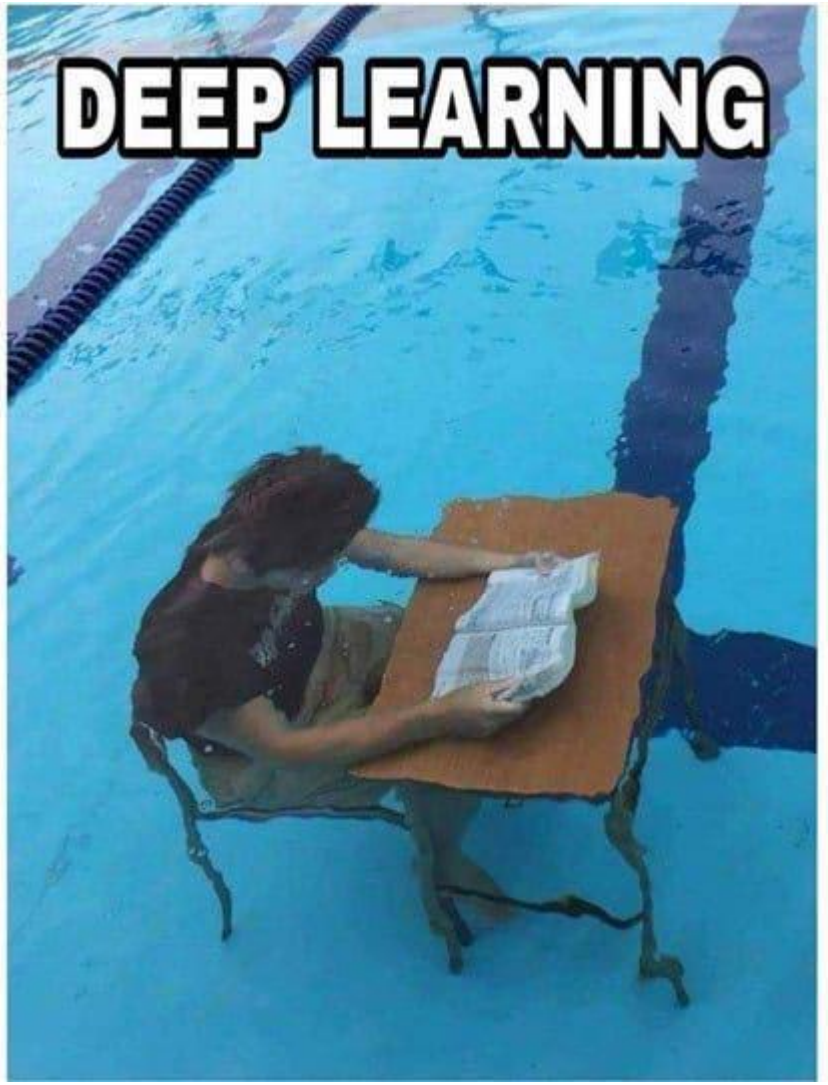
DORN Dawin



VANN Visal

Goals.

- Understand the **fundamental** of deep learning and how it work.
- Learn how **Neural Network** are trained and used in varies **applications**.



Course roadmap

When	Topic
Weeks 1–2	Introduction and neural network foundations
Weeks 3–5	Improving deep neural networks
Weeks 6–8	Convolutional neural networks, architectures, and transfer learning
Week 9	Midterm Exam (MCQ)
Weeks 10–11	Object detection and segmentation, YOLO
Weeks 12–15	RNN, LSTM, attention, transformers, Vision Transformers
Week 16	Final project presentations

Assessment

- Quiz and Class Activity 10%
- Assignment 10%
- Attendant 10%
- Midterm 30%
- Final 40%



Today’s agenda

- What is AI?
- AI vs Machine Learning vs Deep learning
- Type of Machine Learning
- Why Deep Learning Take off?
- Applications of Deep Learning

What is artificial intelligence?

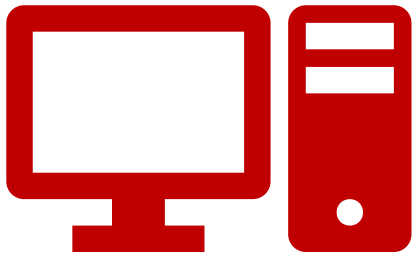
Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

- e.g. computer systems that can perform tasks with human-like intelligence, such as understanding language, recognizing images, learning from data, reasoning, and making decisions.



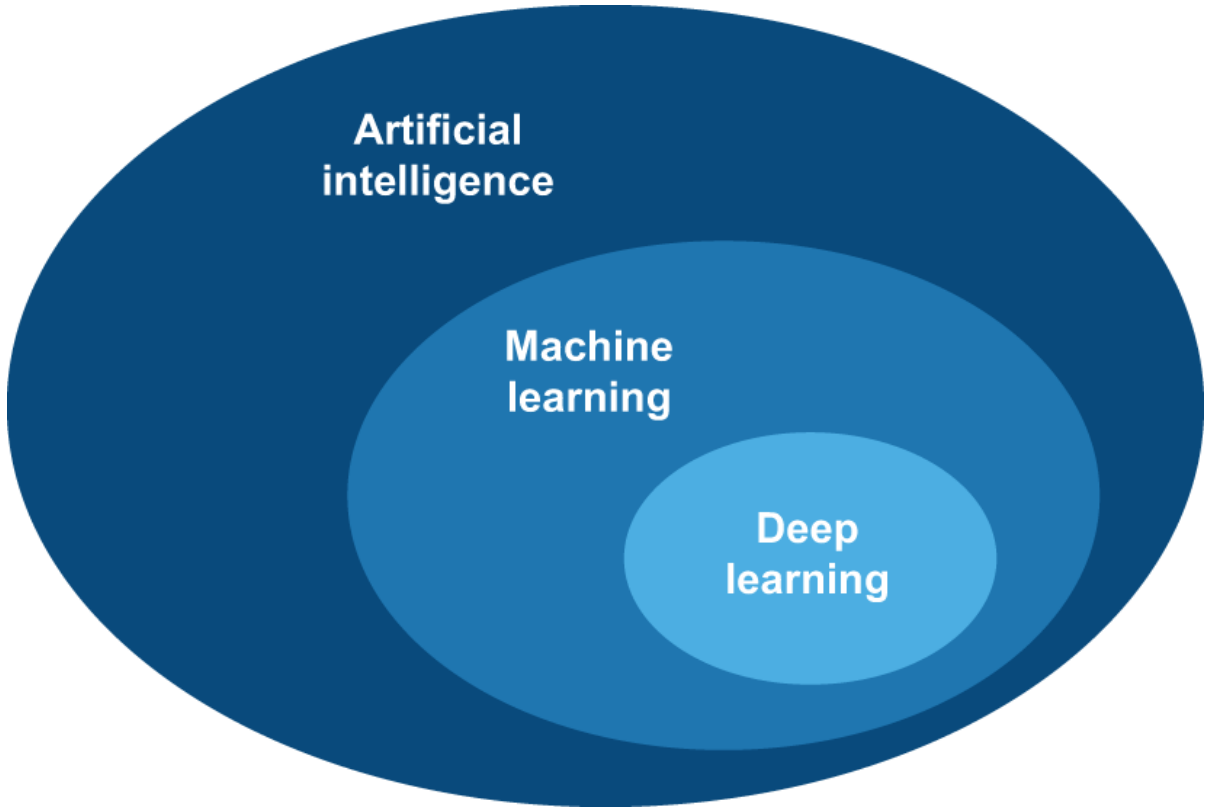
Human

=



AI

AI, machine learning, and deep learning



Artificial intelligence

Any technique that enables computers to mimic human behavior. Example: a chess engine.

Machine learning

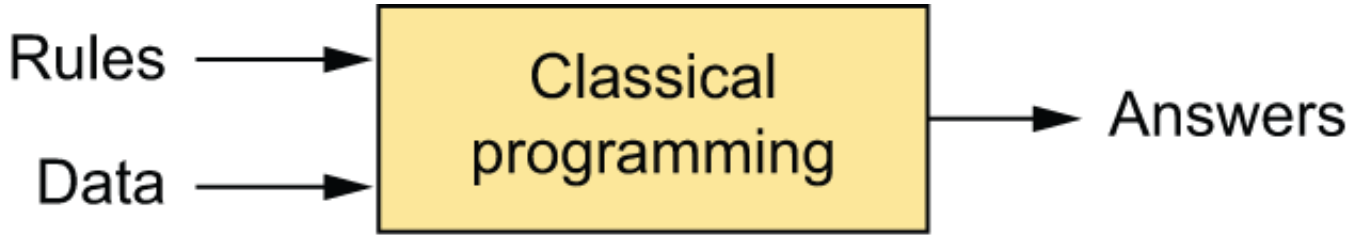
Systems that learn patterns from data instead of following hand-written rules. Example: a spam filter.

Deep learning

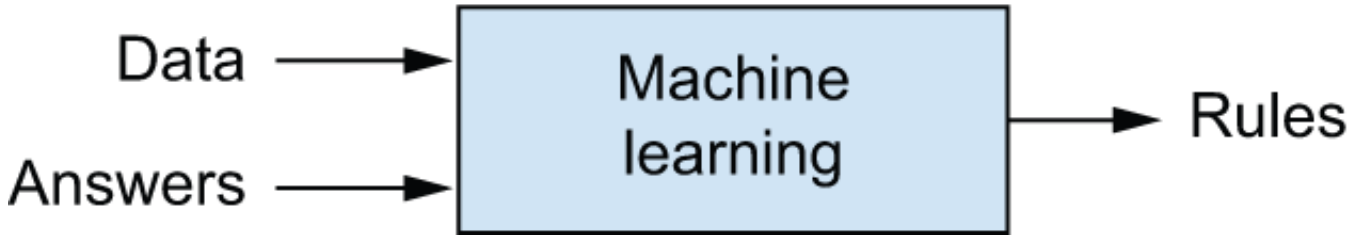
Machine learning with many-layered neural networks that learn their own features from raw data. Example: an image classifier.

Machine learning

Traditional programming



Machine learning



Types of machine learning

Supervised learning

Learn from labeled pairs (x, y).

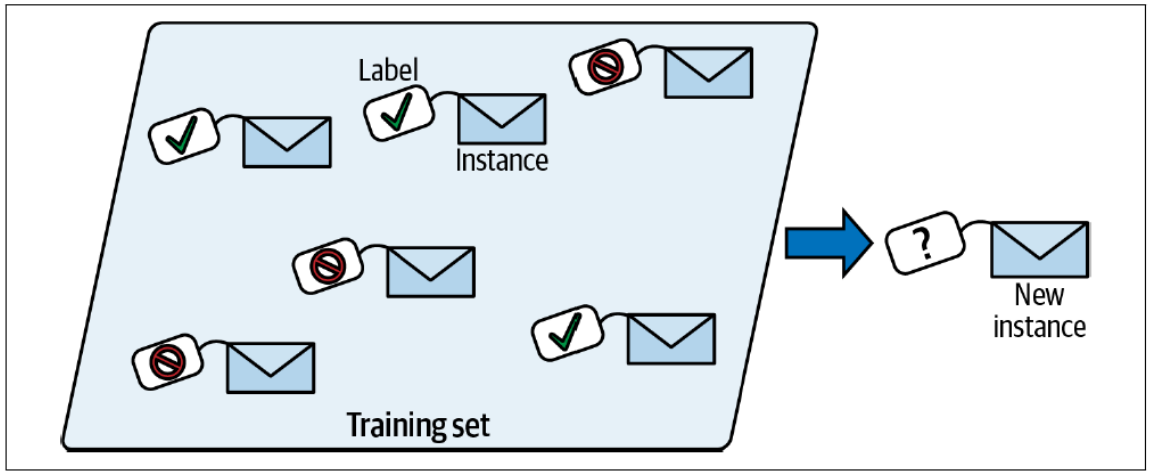


Figure 1-5. A labeled training set for spam classification (an example of supervised learning)

Unsupervised learning

Find hidden patterns in unlabeled data $\{x^{(1)}, .., .. x^{(m)}\}$

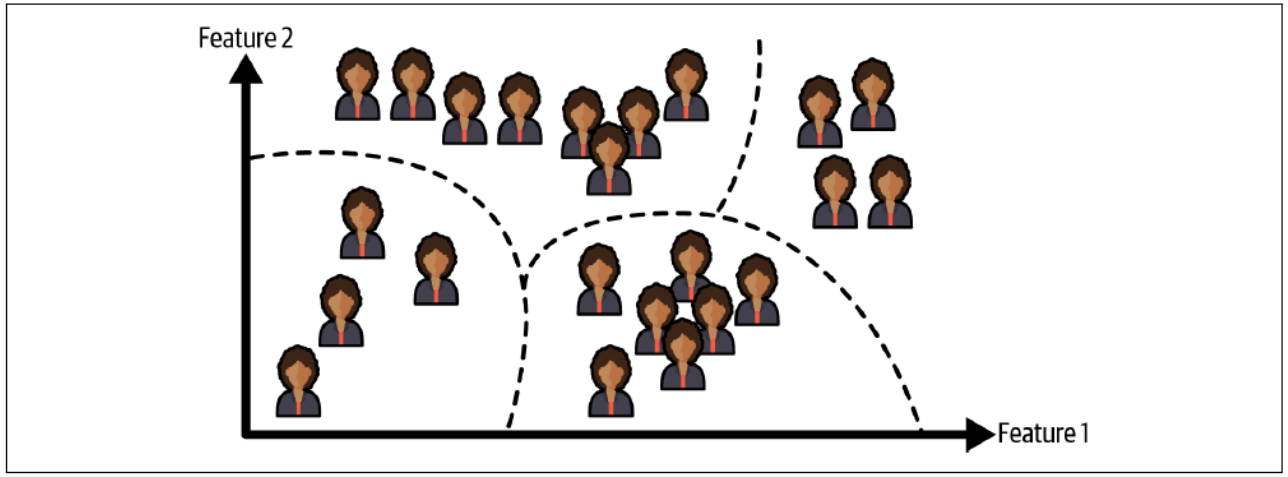


Figure 1-7. Clustering

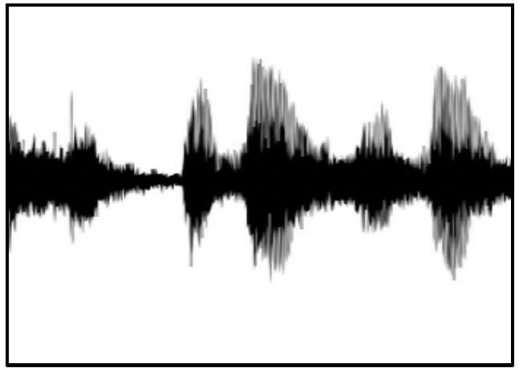
Structured and unstructured data

Structured Data

Size	#bedrooms	….	Price (1000$s)
2104	3		400
1600	3		330
…	…		…
3000	4		540

User Age	Ad Id	….	Click
41	93242		1
80	93287		0
…	…		…
27	71244		1

Unstructured Data



Audio



Image

Four scores and seven years ago…

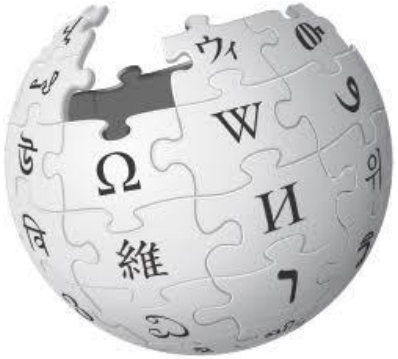
Text

Why deep learning took off?

1. Big Data

- Larger Datasets
- Easier Collection & Storage







2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable



3. Software

- Improved Techniques
- New Models
- Toolboxes

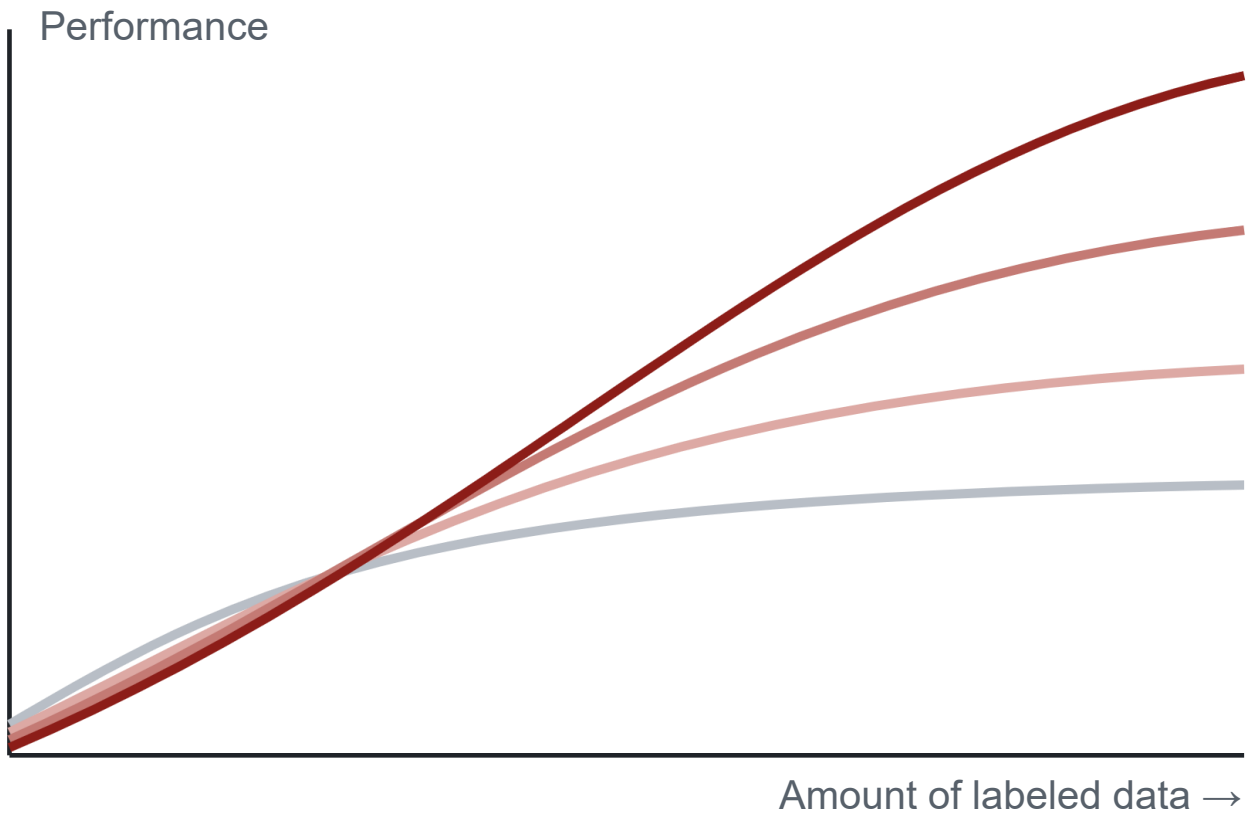








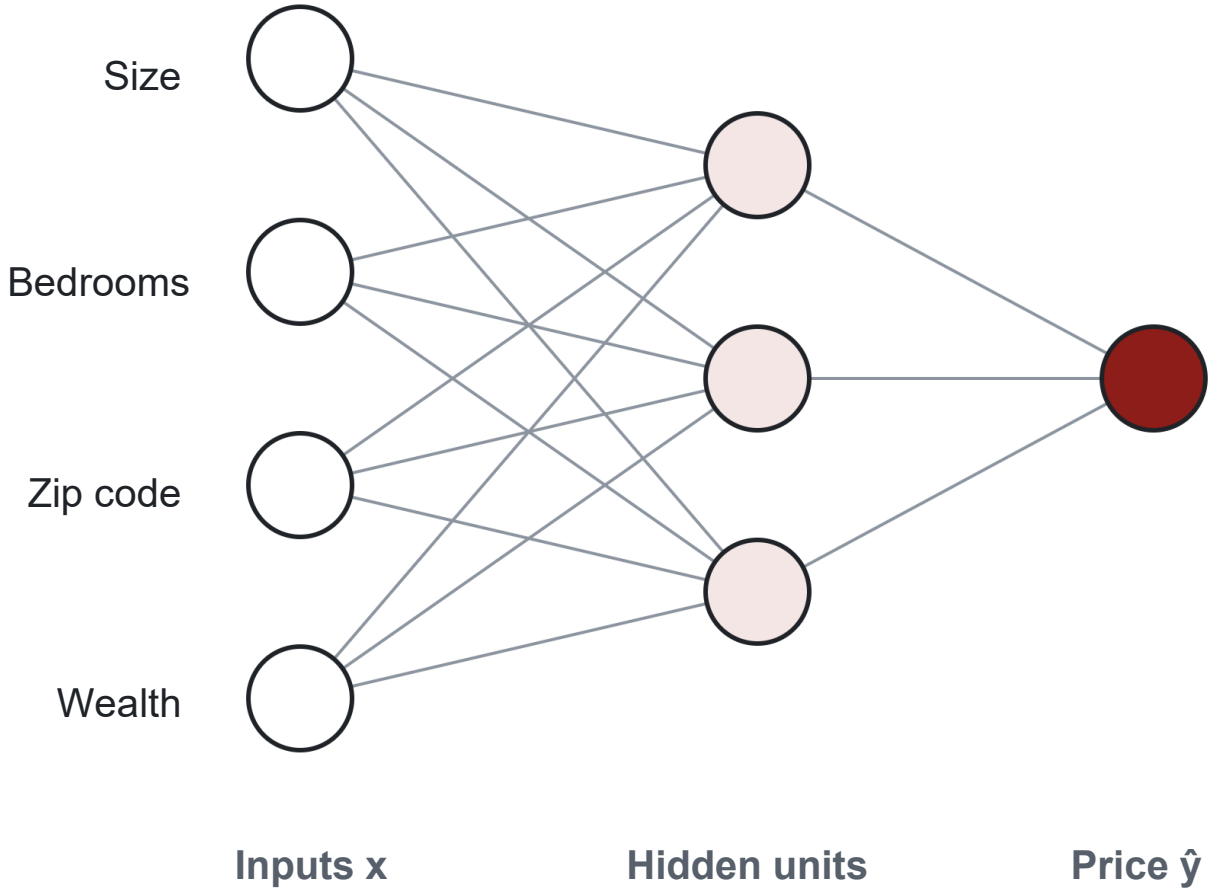
Scale drives deep learning performance



Large neural network
Medium neural network
Small neural network
Traditional ML

Traditional methods plateau. Larger networks keep improving as data grows.

What is a neural network?



Inputs x: features of a house.

Hidden units: intermediate features that the network **learns from data**, not designed by hand.

Output ŷ: the predicted price.

Given enough examples (x, y), training finds a mapping from x to y.

Deep learning around us

Agriculture

Detect crop diseases from photos of leaves; count plants or animals in drone images.

Transport

Count vehicles, check helmet use, and read license plates from traffic cameras.

Healthcare

Screen medical images such as X-rays to support doctors' decisions.

Documents

Digitize printed and handwritten documents with text recognition (OCR), including Khmer script.

Also: speech recognition, translation, recommendation, and chatbots.

Useful materials

- This class is inspired by and is adapted from the Coursera Deep learning course by Andrew Ng (https://www.coursera.org/specializations/deep-learning) and CS231n (https://cs231n.stanford.edu/schedule.html)

- Python Environment
  - Python environment using Anaconda (https://www.anaconda.com/)
  - VsCode text editor (https://code.visualstudio.com/)
  - Google Colab (https://colab.research.google.com/)

- Machine Learning Libraries/Frameworks:
  - Numpy (https://numpy.org/)
  - Scikit-learn (https://scikit-learn.org/)
  - Pytorch (https://pytorch.org)

- Python and Numpy Review (from Standford CS231n by Justin Johnson):
  - http://cs231n.github.io/python-numpy-tutorial/

Useful materials

- YouTube video playlists **Neural Networks: Zero to Hero by Andrej Karpathy**
- Books

